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—Henry Ford

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IoT MQTT Panel app

The IoT MQTT Panel is a highly customisable MQTT-based dashboard application for monitoring real-time telemetry data and remotely controlling IoT devices. The application is available for both Android and iOS platforms. For general setup and operation, refer to the built-in user guide. The dashboard configuration for this system is described in the following figures.

Figures 14 through 17 illustrate the step-by-step configuration of the system dashboard in the IoT MQTT Panel app. Fig. 14 shows how to access the app's user guide and configure the CPU and RAM telemetry panels. Fig. 15 shows the configuration of the RSSI, chip temperature, and flash usage telemetry panels for monitoring the ESP32-C3's operating status. Fig. 16 illustrates the configuration of the last reset reason, Wi-Fi channel, and uptime telemetry panels. Finally, Fig. 17 shows the configuration of the LED control, LED status, and device reset panels, enabling remote control of the onboard LED and remote restarting of the ESP32-C3 directly from the dashboard.

Uploading the firmware and configuring the dashboard

The firmware configuration is now complete. Compile the sketch and upload it to the ESP32-C3 Super Mini by selecting the appropriate board and COM port in the Arduino IDE.

Next, install the IoT MQTT panel app and configure it to connect to the selected MQTT broker. Then, configure the dashboard to display the telemetry data published by the ESP32-C3 Super Mini.

The dashboard displays real-time values for CPU usage, RAM utilisation, Wi-Fi signal strength (RSSI), chip temperature, flash usage, reset reason, device uptime, and Wi-Fi channel. It also provides remote control of the onboard

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LED and supports restarting the ESP32-C3 Super Mini through MQTT commands, as shown in Fig. 18.

Testing

After power-up, the ESP32-C3 Super Mini connects to the configured Wi-Fi network and establishes a connection with the MQTT broker. Once connected, it continuously maintains the MQTT connection, collects system health parameters, publishes telemetry data at one-second intervals, and listens for incoming MQTT commands.

When an ON command is received from the dashboard, the onboard LED turns on. When an OFF command is received, the LED turns off. Similarly, when a RESET command is received, the ESP32-C3 Super Mini restarts.

Throughout operation, the device continuously monitors its health while maintaining low CPU, memory, and network resource usage, making the system suitable for reliable real-time IoT monitoring and control applications. **EFY**



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